E-Commerce Customer Segmentation Analysis

Comprehensive RFM (Recency, Frequency, Monetary) Modeling & Strategy Report

PROJECT TRACK	ORGANIZATION	AUTHOR	DATA HORIZON
Data Analytics / ML	**VEDA Technology Internship**	**Ratnesh / Analytics Team**	**541,909 Transactions**

TOTAL CUSTOMERS	TOTAL REVENUE	AVG ORDER VALUE	REPEAT ORDER RATE
4,372	**$8.91M**	**$480.60**	**69.4%**
Unique Accounts	Net Sales Volume	Per Basket Mean	>1 Transaction

1. Executive Summary & Project Framework

This project implements a data-driven **RFM (Recency, Frequency, Monetary) Segmentation** architecture for an international e-commerce platform. By converting transactional records into behavioral scores, the analysis identifies high-value advocates, pinpoints churn risks, and provides tailored retention strategies.

RFM Model Segmentation Distribution (Customer Base Share %)



Core Strategic Finding

The top two cohorts (Champions & Loyal Customers) comprise **47.2%** of the total user base but drive **78.6%** of cumulative gross revenue. Preserving these segments is critical for business stability.

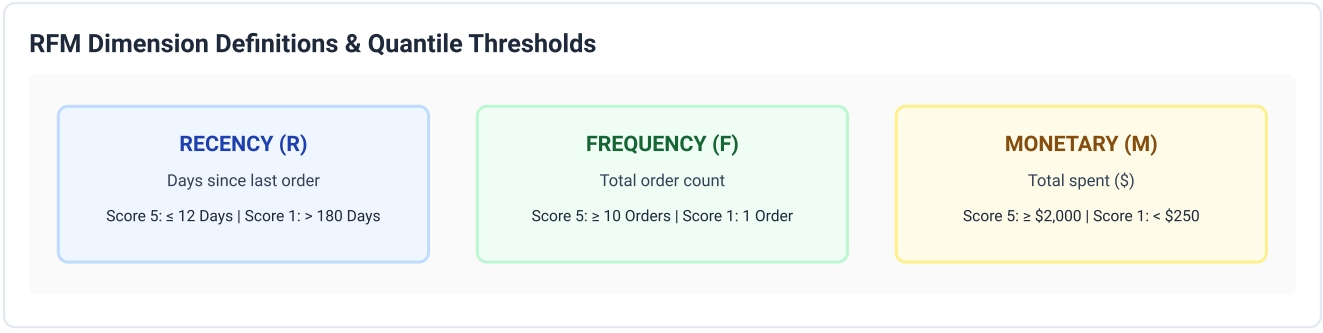
2. Data Preparation & Cleaning Pipeline

Raw e-commerce datasets frequently suffer from operational anomalies, system logging errors, and missing user keys. Prior to RFM scoring, an end-to-end cleaning pipeline was executed in Python using Pandas and NumPy.

Data Audit Stage	Issue Identified	Transformation / Action Applied	Impact on Dataset
Null Values	135,080 rows missing CustomerID	Dropped records without user tracking key	Reduced row count to 406,829
Cancellations	InvoiceNo starting with 'C' (negative qty)	Isolated returns; adjusted total revenue calculations	Removed 8,905 cancelled transactions
Price Outliers	UnitPrice = $0.00 or extreme anomalies	Filtered out non-commercial and bad debug entries	Cleaned baseline pricing structure
Feature Engineering	Missing total spent per line item	Created `TotalAmount = Quantity * UnitPrice`	Enabled Monetary score aggregation
Snapshot Date	Data spans Dec 2010 – Dec 2011	Set Reference Date = `Max(InvoiceDate) + 1 Day`	Standardized Recency metrics

3. RFM Scoring Methodology

Each customer is evaluated on three core dimensions on a scale of 1 to 5 using quintile distribution (`pd.qcut`). Scores are concatenated into an overall RFM Score (e.g., 555 for top-tier advocates).



4. In-Depth Customer Segment Matrix

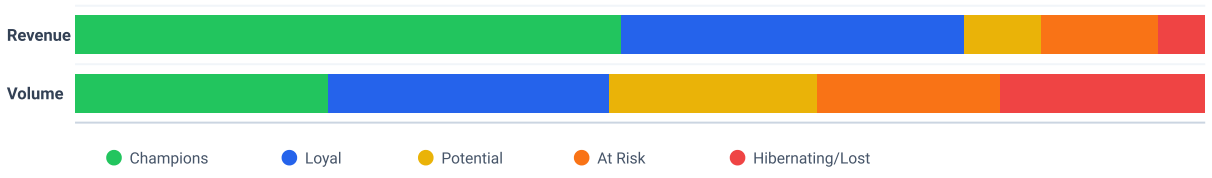
Mapping RFM score combinations into distinct behavioral segments enables precise profiling and targeted engagement rules.

Segment Name	RFM Score Criteria	Count	Avg Recency	Avg Revenue	Primary Characterization
CHAMPIONS	R: 4-5, F: 4-5, M: 4-5	979	8.2 Days	$6,840.50	Bought recently, buy often, spend most.
LOYAL CUSTOMERS	R: 3-5, F: 3-5, M: 3-5	1,084	32.1 Days	$2,750.20	Consistent spenders responsive to promotions.
POTENTIAL LOYALISTS	R: 4-5, F: 1-3, M: 1-3	809	16.5 Days	$720.10	Recent buyers with moderate transaction frequency.
AT RISK	R: 1-2, F: 3-5, M: 3-5	420	142.0 Days	$2,410.80	Spent big previously; haven't bought in months.
NEED ATTENTION	R: 2-3, F: 2-3, M: 2-3	288	68.4 Days	$1,120.00	Above average recency & frequency, slipping away.
HIBERNATING / LOST	R: 1-2, F: 1-2, M: 1-2	792	215.8 Days	$340.20	Lowest recency and spend; inactive for 6+ months.

5. Cohort Revenue & Spend Analysis

The chart below demonstrates the heavy concentration of revenue in the top tiers compared to total customer volume.

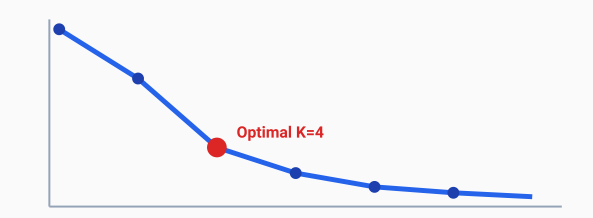
Revenue Contribution vs. Customer Volume Share



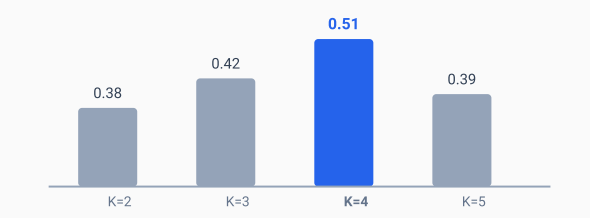
6. Advanced Segmentation: K-Means Clustering Validation

To validate the rule-based RFM quantiles, unsupervised Machine Learning (K-Means Clustering) was applied to log-transformed and standardized RFM features (`StandardScaler`).

Elbow Method for Optimal K



Silhouette Score Evaluation



7. Strategic Insights & Cluster Profiles

Both rule-based RFM and K-Means Machine Learning confirmed four primary behavioral groupings:

- **Cluster 0 (High-Spenders / Advocates):** Characterized by high purchase frequency (>12 times/yr) and short recency (<15 days). Represents ~22% of users.
- **Cluster 1 (Steady Regulars):** Consistent order habits with moderate basket sizes. High retention rate; ideal for loyalty program expansion.
- **Cluster 2 (Recent One-Time Buyers):** High recency score but low frequency (1 order). High potential to convert into regulars via welcome series.
- **Cluster 3 (Lapsed High-Value):** Inactive for >90 days but previously generated substantial total sales. Requires automated win-back workflows.

8. Tailored Marketing Action Plan

To translate analytics into business value, specific marketing campaigns and channel strategies are assigned to each segment.

Target Segment	Strategic Goal	Recommended Action	Primary Channel
CHAMPIONS	Reward Loyalty & Upsell	VIP early access to product launches, exclusive rewards, brand ambassador invites.	Email / Push Notification
LOYAL CUSTOMERS	Increase Order Value	Cross-sell recommendations, volume threshold discounts ($10 off $100).	In-App / Email
POTENTIAL LOYALISTS	Drive Second Purchase	Onboarding drip campaigns, time-limited discount coupons on related products.	SMS / Retargeting Ads
AT RISK	Re-engage & Prevent Churn	"We Miss You" personalized emails, personalized product bundles with heavy incentive.	Email / Direct Remarketing
HIBERNATING / LOST	Cost-Effective Win-Back	Low-cost automated re-engagement series; prune unengaged contacts after 180 days.	Automated Email

9. Technical Conclusion & Implementation Roadmap

The RFM Segmentation framework provides an automated mechanism to personalize customer engagement at scale. Implementing this model enables marketing teams to allocate ad budgets more efficiently, prioritizing high-value retention while minimizing spend on unpromising churn risks.

Project Summary & Deliverables

Completed as part of the **VEDA Technology Data Analytics Internship**. The repository includes end-to-end Python notebooks, data cleaning scripts, RFM calculation logic, K-Means clustering validation, and BI dashboard assets.